WMHV

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Document details

Categories of variables	Imaging outcome measure – Insight46 Phase 2 cross-sectional global white matter hyperintensity (WMH) volume (WMHV) generated using the cross-sectional Bayesian Model Selection (BaMoS) pipeline.
Summary of work undertaken	Cleaning± Description/Rationale: Phase 2 cross-sectional WMHV data was generated by Carole Sudre using the BaMoS pipeline. The WMH segmentations were then QCed by Dr Carole Sudre and Prof. Jo Barnes

BaMoS outcome variables Phase 2 Insight 46

This document details all variables created from the Insight46 Phase 2 cross-sectional Bayesian Model Selection (BaMoS) pipeline (Insight46 phase 1 and phase 2), how they were cleaned and quality controlled (QC), and recommendations for use.

Global Phase 2 cross-sectional BaMoS outcome variables

Phase 2 WMH masks were generated from Phase 2 FLAIR images by Dr Carole Sudre using BaMoS (Sudre et al., 2015). Global WMHV were generated from all available masks, and are expressed in millilitres.

Longitudinal WMH mask QC and inclusion criteria

QC of all Phase 2 WMH masks was performed by Prof. Jo Barnes and Dr Carole Sudre by visually inspecting masks overlaid on corresponding FLAIR images. Masks with segmentation issues were re-run with modified scripts and corrected. All corrected masks were reviewed by Prof. Jo Barnes in order to determine if the corrections had solved the segmentation issues and were appropriate for use. All masks passed QC once corrected. Individuals who had evidence of cortical stroke (either self-/clinically-reported, reported by radiologists or identified during QC) are removed from the dataset. This is because hyperintense signals, known to be caused by other conditions, influence the volume of WMH of presumed vascular origin (age-related cerebral small vessel disease). In addition, not all stroke lesions were consistently segmented, giving variable levels of stroke lesion inclusion/exclusion in the masks between subjects. Individuals with other neurological disorders were not otherwise excluded at this stage (see "Recommended Complimentary Datasets").

Recommendations for Use

This data is intended for Phase 2 cross-sectional analysis only, and should not be used in combination with Phase 1 cross-sectional WMHV data to calculate change in WMHV. For longitudinal analysis of WMHV change between Phase 1 and Phase 2, we recommend using **bamos_wmhvc_i46p2** which was produced from a different longitudinal pipeline.

Recommended Complimentary Datasets

Insight46 MRI QC Summary Phase 2 - Use to determine T1, T2 and FLAIR QC pass/fail for use in combination with other imaging data. - Source: library file: **I46P2_neuropsychology** on Condor - Data sheet: Search for the library file above and select any variable. The document will be linked in the variable level metadata page

Clinical Diagnosis of Neurological Disorders Phase 2

In this longitudinal WMH volume dataset we have excluded all subjects with any evidence of cortical stroke indicated with a "Y" in the "Stroke_DONOTUSE_if_Y" column, as mentioned above. Depending on intended analysis, researchers may also wish to exclude other neurological conditions that have been shown to influence WMH volume changes.

- Library file: **I46P2_neuropsychology** on Condor
- Data sheet: Search for the library file above and select any variable. The document will be linked in the variable level metadata page

Total Intracranial Volume - It is recommended that analyses using WMH data are adjusted for TIV. - Source: The variable **spm_tiv_vol_l_i46p2** on Condor which is in the library file: **I46P2_brainVolumes_L**

Variables:

subject: Insight46 unique subject identifier **globalwmhv_i46p2**: Global white matter hyperintensity (WMH) volume - Cross-sectional - Insight46 phase 2 **globalwmhvs_i46p2**: Phase 2 cross-sectional global WMHV for the supratentorial region (brainstem and cerebellum removed) only

References

Sudre, C.H. et al. (2015) 'Bayesian model selection for pathological neuroimaging data applied to white matter lesion segmentation.', IEEE transactions on medical imaging, 34(10), pp. 2079–2102. Available at: https://doi.org/10.1109/TMI.2015.2419072.

Variables in this document

5 medium-confidence matches

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Heart and circulatory conditions [51014]		
stroke	Have you had a stroke in the last 10 years - at age 53 years	prose scan
Sub-Studies [57] › Insight46 [699] › Brain MRI [6033] › Brain volumes [60331] › Brain volumes - Longitudinal (phase 1 and 2) [603312]		
spm_tiv_vol_l_i46p2	Total intracranial volume (cm3) - Longitudinal (phase 1 and 2 only) - Insight46 phase 2	prose scan
Sub-Studies [57] › Insight46 [699] › Brain MRI [6033] › White matter hyperintsity (WMH) volumes [60332] › White matter hyperintsity (WMH) volumes - Cross-sectional [603321]		
globalwmhv_i46p2	Global white matter hyperintensity (WMH) volume - Cross-sectional - Insight46 phase 2	prose scan
globalwmhvs_i46p2	Global white matter hyperintensity (WMH) volume supratentorial region (brainstem and cerebellum removed) only - Cross-sectional - Insight46 phase 2	prose scan

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Brain MRI [6033] › White matter hyperintsity (WMH) volumes [60332] › White matter hyperintsity (WMH) volumes - Longitudinal (phase 1 and 2) [603322]		
bamos_wmhvc_i46p2	Change in white matter hyperintensity (WMH) volume from baseline to follow-up (p2_bamos_wmhv_long – p1_bamos_wmhv_long) - Longitudinal (phase 1 and 2 only) - Insight46 phase 2	prose scan